

Quarto Story Telling

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Storytelling in R

Introduction

Data interpretation and storytelling mean understanding the results of data analysis, answering important questions, and sharing the findings with others. Visual tools like graphs, charts, plots, and dashboards are often used to make the information clearer and easier to explain. In this chapter, we will look at different data analysis tools and techniques, as well as the different types of data analytics.

A version of this textbook is available at <https://nxrolinf.github.io/QuartoSTell>

Lab 3: Data Storytelling and Visualization in R

Overview

This lab explores data storytelling using R through: - Exploratory data analysis (EDA) - Transformations and skewness - Regression visualization - Time series anomalies - Composite plots - Interactive visualization (Shiny) - A capstone narrative

Setup

```
# Install if needed
# install.packages(c("tidyverse", "moderndive", "fivethirtyeight",
#                   "patchwork", "scales", "shiny"))

library(tidyverse)
library(moderndive)
library(fivethirtyeight)
library(patchwork)
library(scales)
```

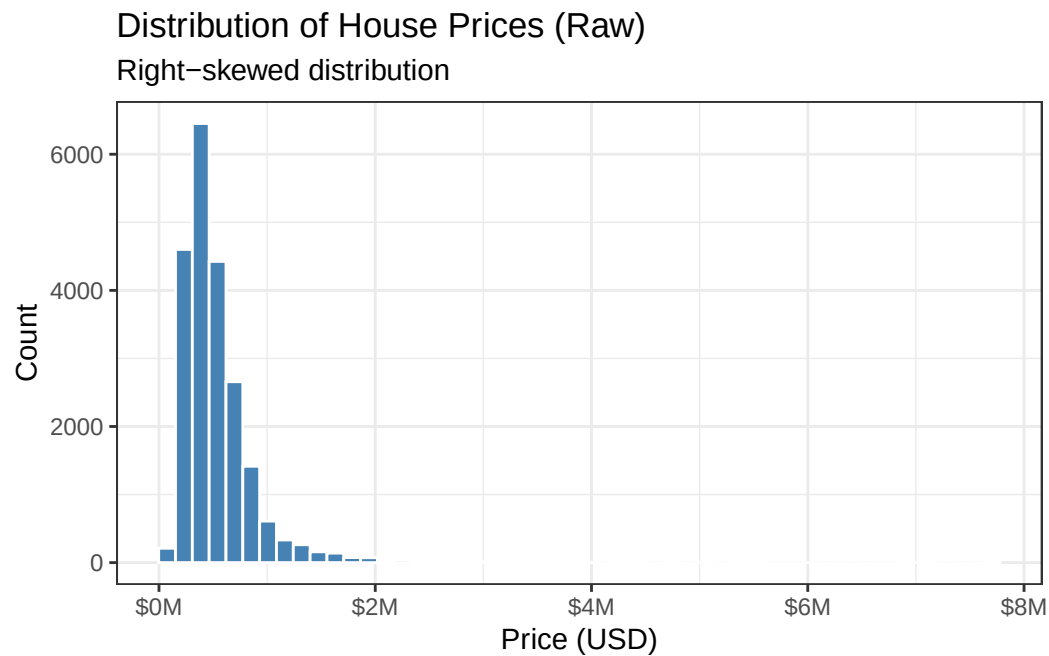
Exercise 1: Exploratory Data Analysis & Skewness

Raw Distribution

```
data("house_prices")

p1_raw <- ggplot(house_prices, aes(x = price)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white") +
  scale_x_continuous(labels = label_dollar(scale = 1e-6, suffix = "M")) +
  labs(
    title = "Distribution of House Prices (Raw)",
    subtitle = "Right-skewed distribution",
    x = "Price (USD)",
    y = "Count"
  ) +
  theme_bw()

p1_raw
```



Log Transformation

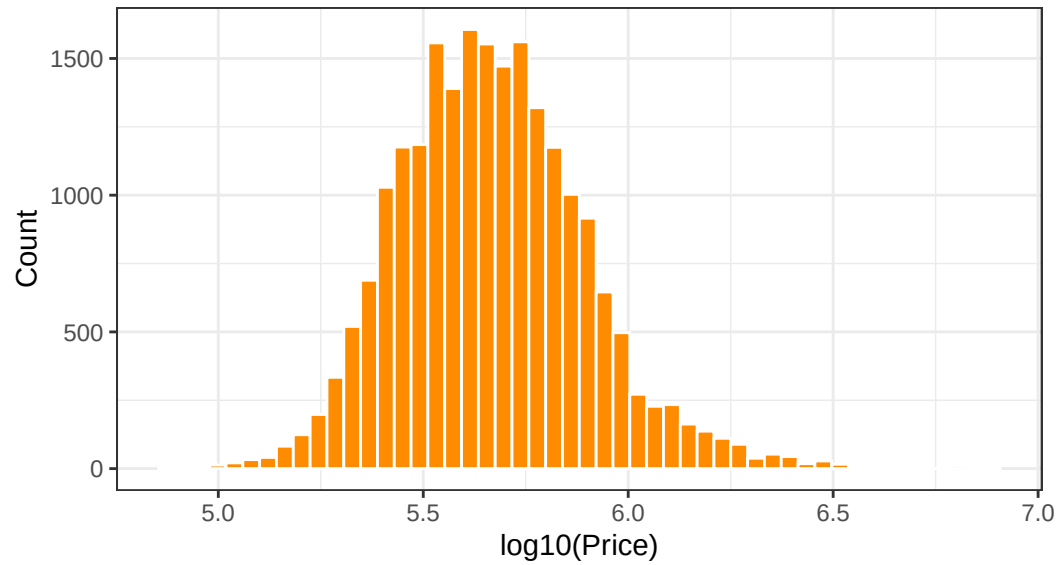
```
house_prices <- house_prices %>%
  mutate(
    log10_price = log10(price),
    log10_size = log10(sqft_living)
  )

p1_log <- ggplot(house_prices, aes(x = log10_price)) +
  geom_histogram(bins = 50, fill = "darkorange", color = "white") +
  labs(
    title = "Distribution of House Prices (log10 scale)",
    subtitle = "More symmetric distribution",
    x = "log10(Price)",
    y = "Count"
  ) +
  theme_bw()

p1_log
```

Distribution of House Prices (log10 scale)

More symmetric distribution



Interpretation

The raw distribution is heavily right-skewed due to high-value outliers.

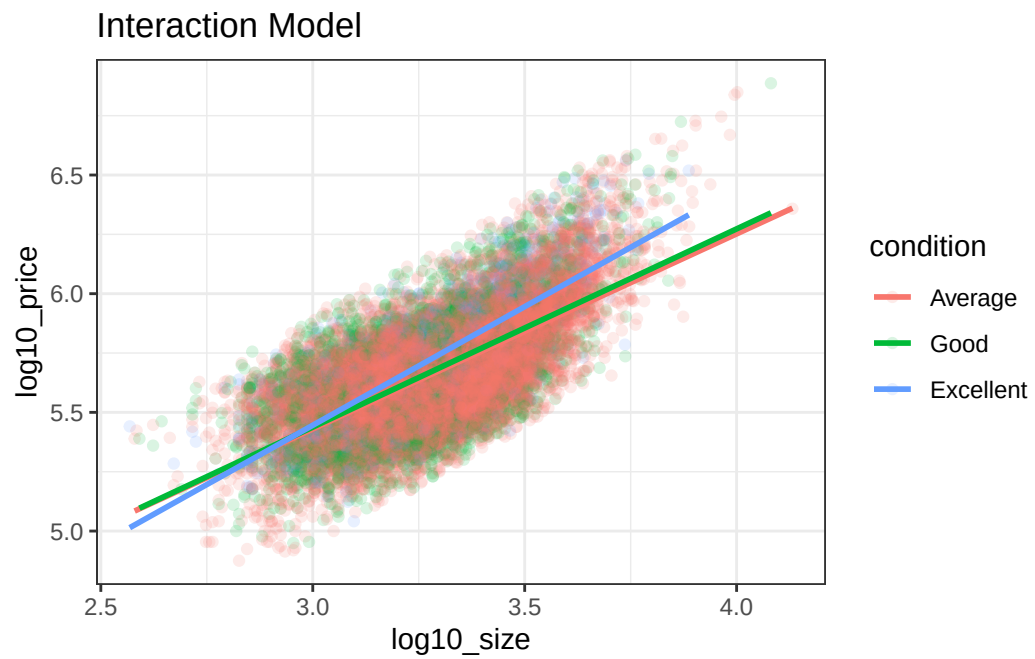
The log transformation reveals a more symmetric, near-normal distribution, improving interpretability.

Exercise 2: Interaction vs Parallel Slopes

```
house_prices <- house_prices %>%  
  mutate(condition = factor(condition, levels = 1:5,  
                             labels = c("Poor", "Fair", "Average", "Good", "Excellent")))  
  
house_sub <- house_prices %>%  
  filter(condition %in% c("Average", "Good", "Excellent"))
```

Interaction Model

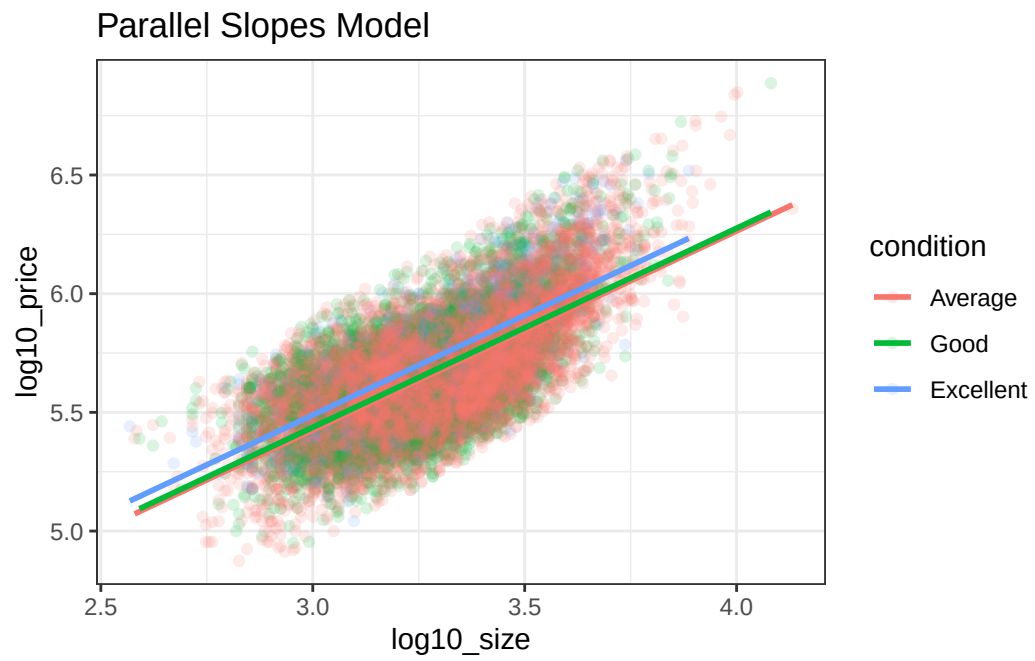
```
p2_interaction <- ggplot(house_sub,  
                          aes(x = log10_size, y = log10_price, color = condition)) +  
  geom_point(alpha = 0.15) +  
  geom_smooth(method = "lm", se = FALSE) +  
  labs(title = "Interaction Model") +  
  theme_bw()  
  
p2_interaction
```



Parallel Slopes Model

```
p2_parallel <- ggplot(house_sub,
                      aes(x = log10_size, y = log10_price, color = condition)) +
  geom_point(alpha = 0.15) +
  moderndive::geom_parallel_slopes(se = FALSE) +
  labs(title = "Parallel Slopes Model") +
  theme_bw()

p2_parallel
```



Interpretation

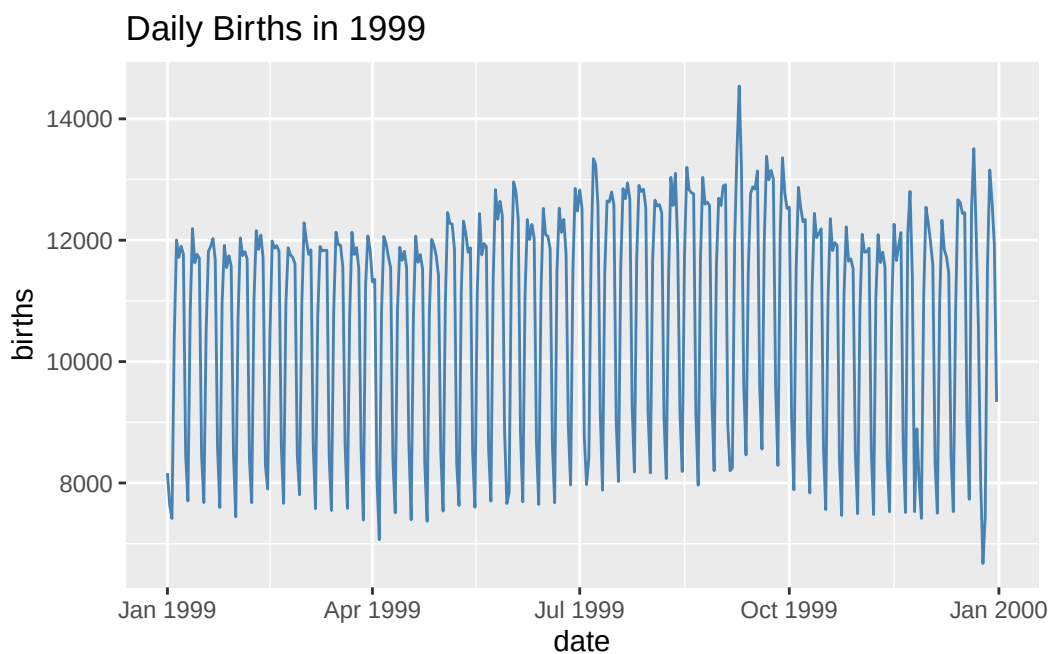
Slopes are similar across conditions, supporting a simpler parallel slopes interpretation: - Size affects price consistently - Condition shifts baseline price

Exercise 3: Time Series Anomaly

```
data("US_births_1994_2003")

births_1999 <- US_births_1994_2003 %>%
  filter(year == 1999) %>%
  mutate(date = as.Date(paste(year, month, date_of_month, sep = "-")))

ggplot(births_1999, aes(x = date, y = births)) +
  geom_line(color = "steelblue") +
  labs(title = "Daily Births in 1999")
```



Peak Detection

```
births_1999 %>%
  arrange(desc(births)) %>%
  head()
```

Peak Detection

```
# A tibble: 6 x 6
  year month date_of_month date      day_of_week births
<int> <int>      <int> <date>      <ord>      <int>
1  1999     9          9 1999-09-09 Thurs      14540
2  1999    12         21 1999-12-21 Tues      13508
3  1999     9          8 1999-09-08 Wed       13437
4  1999     9         21 1999-09-21 Tues      13384
5  1999     9         28 1999-09-28 Tues      13358
6  1999     7          7 1999-07-07 Wed       13343
```

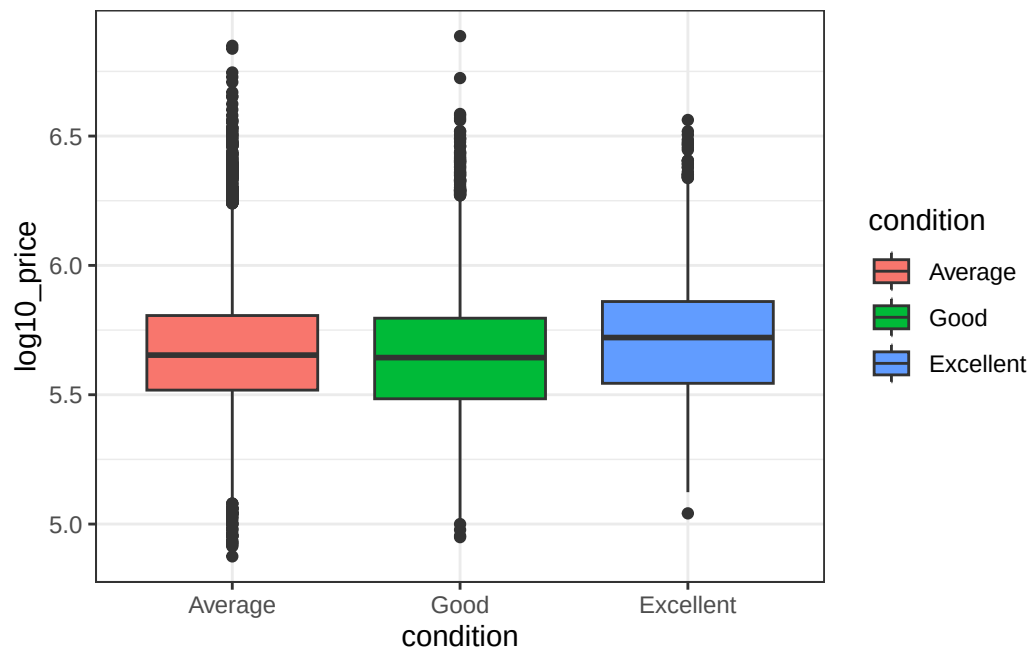
Interpretation

The largest spike occurs on **September 9, 1999 (9/9/99)**, likely due to cultural preferences influencing scheduled births.

Exercise 4: Composite Visualization

Plot A

```
p4a <- ggplot(house_sub,
              aes(x = condition, y = log10_price, fill = condition)) +
  geom_boxplot() +
  theme_bw()
p4a
```

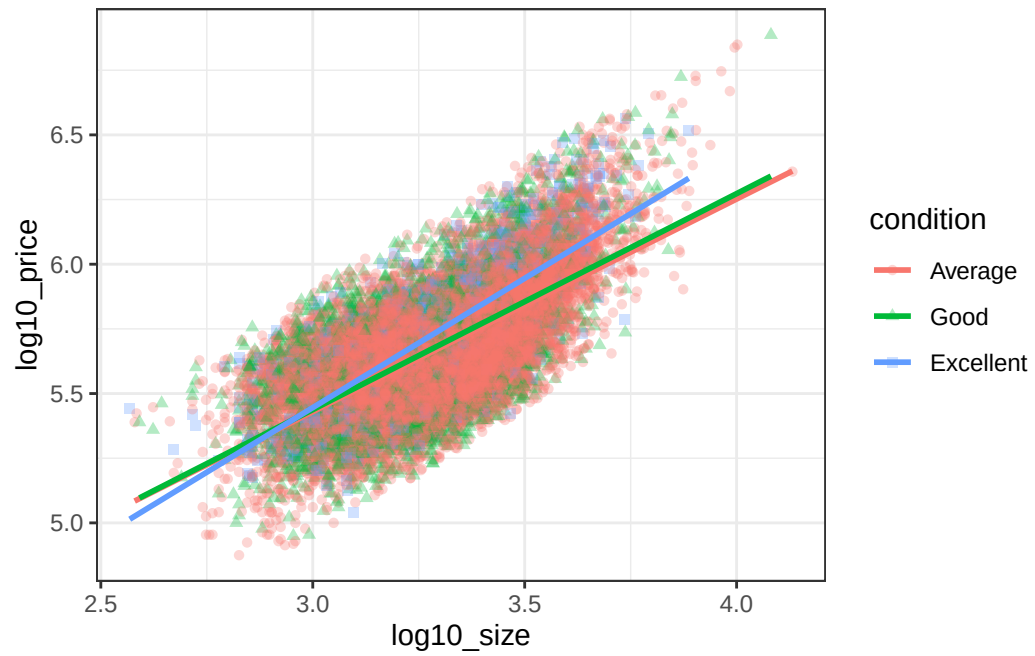


Plot B

```
p4b <- ggplot(house_sub,
              aes(x = log10_size, y = log10_price,
                  color = condition, shape = condition)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE) +
```

Exercise 4: Composite Visualization

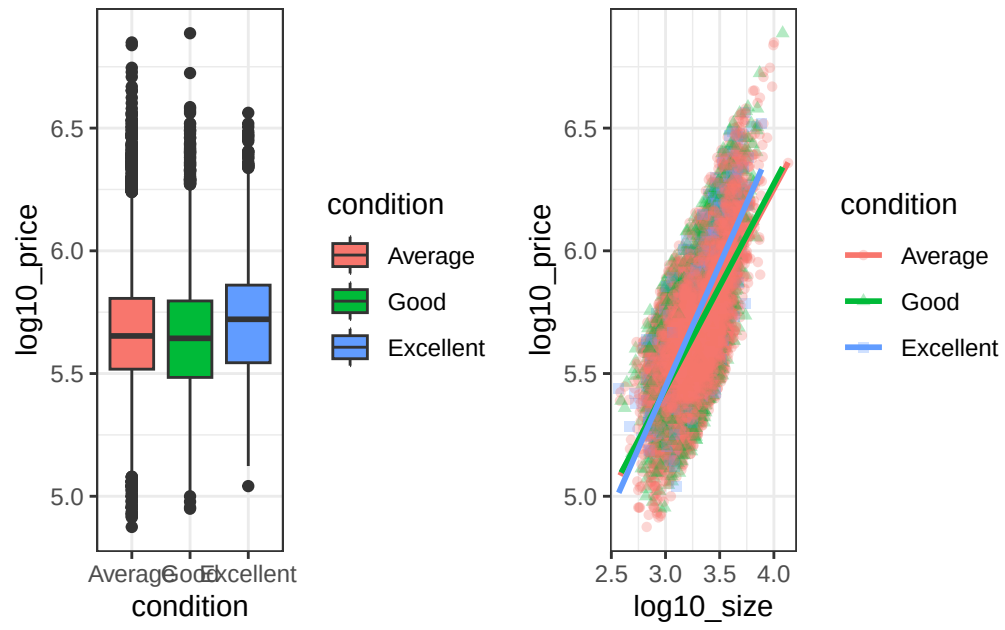
```
theme_bw()  
p4b
```



Combined

```
p4a | p4b
```

Exercise 4: Composite Visualization



Exercise 5: Shiny App

Run separately as `app.R`

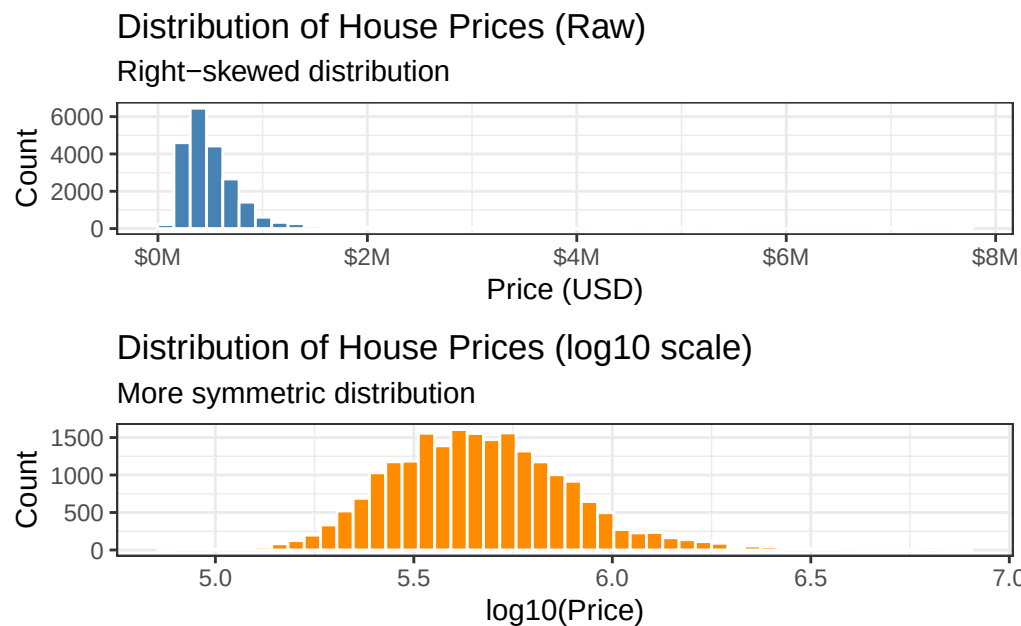
```
# shinyApp(ui, server)
```

(App code omitted in rendering but included in source file)

Capstone: What Drives House Prices?

Step 1: Distribution

```
cap1 <- p1_raw / p1_log  
cap1
```



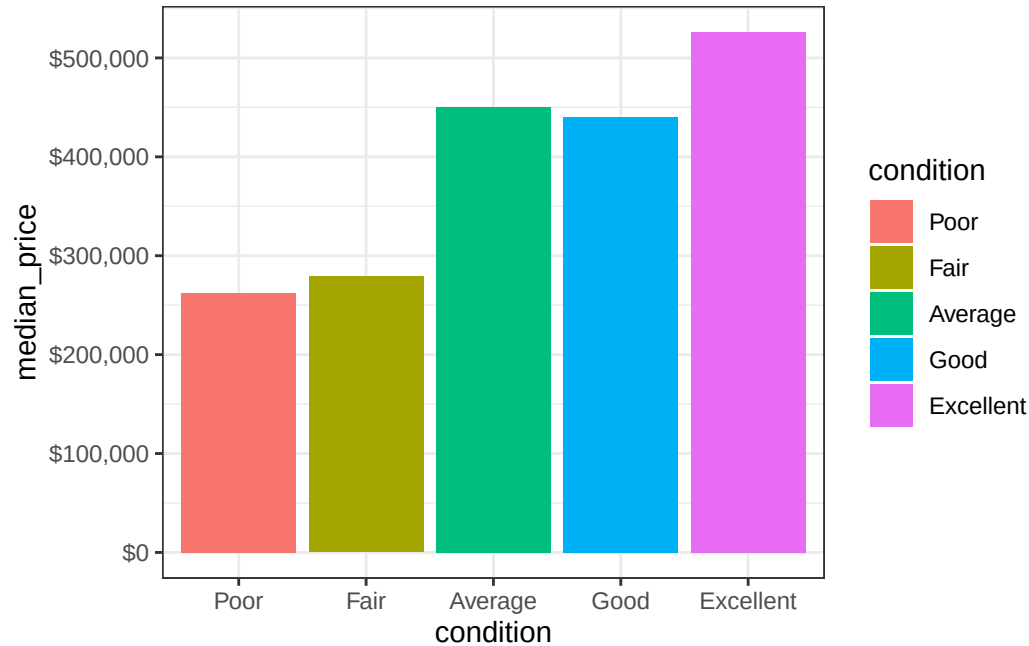
Step 2: Condition Premium

```
median_by_cond <- house_prices %>%  
  group_by(condition) %>%  
  summarise(median_price = median(price), .groups = "drop")  
  
cap2 <- ggplot(median_by_cond,  
               aes(x = condition, y = median_price, fill = condition)) +  
  geom_col() +
```

Step 3: Size + Condition

```
scale_y_continuous(labels = label_dollar()) +  
theme_bw()
```

cap2

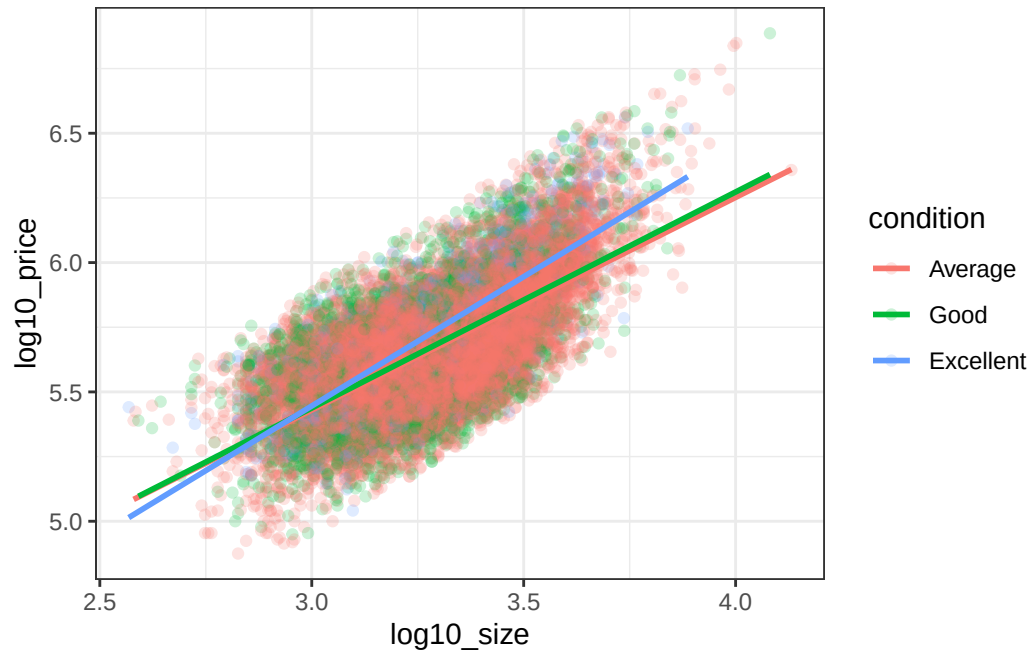


Step 3: Size + Condition

```
cap3 <- ggplot(house_sub,  
               aes(x = log10_size, y = log10_price, color = condition)) +  
  geom_point(alpha = 0.2) +  
  geom_smooth(method = "lm", se = FALSE) +  
  theme_bw()
```

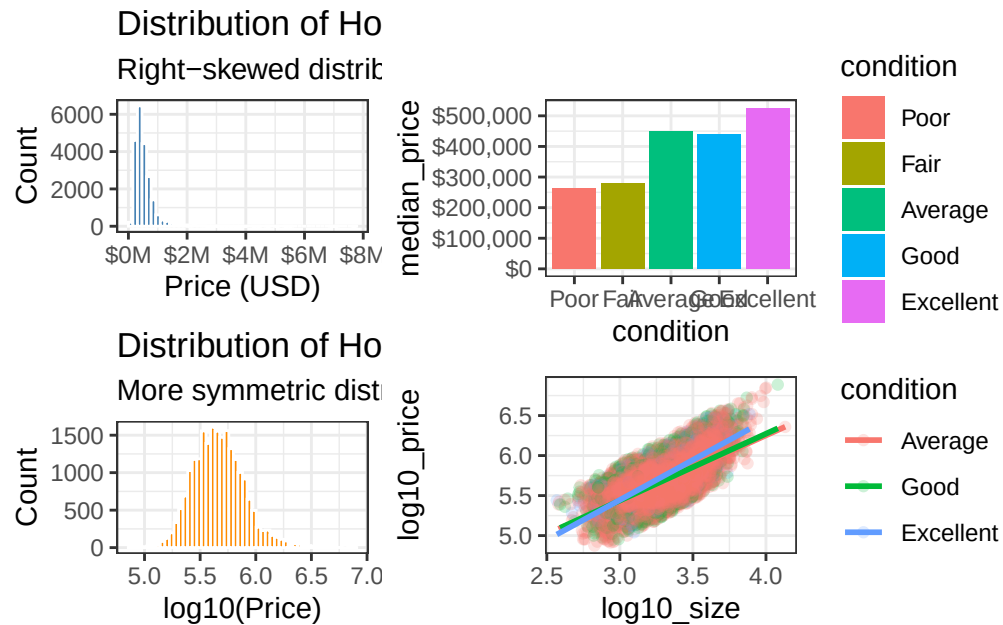
cap3

Final Narrative



Final Narrative

cap1 | (cap2 / cap3)



Conclusions

- House prices are **right-skewed**, requiring transformation.
- **Condition strongly affects price levels.**
- **Size increases price consistently across conditions.**

Call to Action

- Sellers should invest in condition improvements.
 - Buyers can find value in larger homes needing upgrades.
-

Appendix: Shiny App Code

```
#!/ eval: false
library(shiny)

house_shiny <- house_prices %>%
  filter(condition %in% c("Average", "Good", "Excellent"))

ui <- fluidPage(
  titlePanel("House Prices Explorer"),
  sidebarLayout(
    sidebarPanel(
      selectInput("condition_select", "Condition:",
                  choices = c("Average", "Good", "Excellent"))
    ),
    mainPanel(plotOutput("scatter_plot"))
  )
)

server <- function(input, output) {
  filtered_data <- reactive({
    house_shiny %>% filter(condition == input$condition_select)
  })

  output$scatter_plot <- renderPlot({
    ggplot(filtered_data(),
            aes(x = log10_size, y = log10_price)) +
    geom_point(alpha = 0.3) +
    geom_smooth(method = "lm") +
    theme_bw()
  })
}

shinyApp(ui, server)
```

! Work task

For the final part of the lab, you must produce a coherent data story that includes:

1. **At least three different plot types.**
2. **Proper annotation** (titles, axis labels with units, and captions).
3. **A call to action** or a concluding summary of what the visuals reveal.

! Delivery of your task

You're supposed to prepare a qmd document (Quarto, have to install the Quarto CLI from quarto.org). This is a lightweight notebook, fully in R.

Use the **visual** interface for easier development (check the small toolbar, on top of your document window):

1. New file -> Quarto document (you can use any format, as you'll deliver the qmd file), just check that the R kernel is enabled.
2. Organize your document into sections/lists, like any simple HTML document.
3. Add your R code, and execute it (insert Executable cell, R), to make sure that you have all the required libraries. If needed, include the necessary libraries at the top of your file, as an executable cell.
4. Render, and you'll check the output in your browser.
5. Notice that you can also include simple code, or inhibit code execution (check various insert options, and options for the various cells)